Time-Reversal Symmetry in Transformer Predictions on the Cylinder Graph HMM

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1 Introduction

A natural question in the study of transformers trained on Hidden Markov Model (HMM) sequences is whether the *direction* of time affects learnability. Information-theoretically, the entropy rate of a stationary process is invariant under time reversal: the forward and backward sequences contain the same amount of uncertainty per token. A Bayes-optimal predictor with infinite context therefore achieves the same cross-entropy loss in both directions.

However, a finite-context transformer may find one direction systematically easier if the latent belief state converges to its stationary distribution at a different rate in each temporal direction. The convergence rate depends on the spectral properties of the transition matrix, which is generally different for the forward and reverse chains.

This report investigates that question on the Cylinder Graph HMM (fully described in `notes/model_comparison` whose directed transition structure creates an a priori asymmetry between the forward and reverse processes. We formulate seven testable hypotheses (Section 3), describe the experimental setup (Section 4), present theoretical predictions including the Bayes-optimal gap at context length $k = 16$ (Section 5), and compare against a matched architecture sweep (Section 6). We also report complementary experiments on belief-state localisation via activation ablations (Section 7), on the structure of the learned token-embedding geometry (Section 8), and on forward–reversed training dynamics (Section 9).

2 Theoretical Background

2.1 Entropy Rate Invariance

For a stationary stochastic process $\{X_t\}$, the entropy rate is

$$h \;=\; \lim_{n\to\infty} \frac{1}{n}\, H(X_1, \ldots, X_n). \tag{1}$$

Because the joint entropy $H(X_1, \ldots, X_n)$ depends only on the joint distribution and not on the ordering of its arguments, the entropy rate of the forward process $(\ldots, X_{t-1}, X_t, X_{t+1}, \ldots)$ is identical to that of the time-reversed process $(\ldots, X_{t+1}, X_t, X_{t-1}, \ldots)$.

Consequence. Any difference in transformer loss between forward and backward prediction cannot be explained by an inherent information-theoretic asymmetry. The Bayes-optimal predictor with infinite context achieves identical loss in both directions.

2.2 Reverse HMM Construction

Given a forward HMM with joint emission–transition matrices $E[j,i,k] = P(x_t{=}j,\ s_{t+1}{=}k \mid s_t{=}i)$ and stationary distribution π, the reverse-time HMM has emission matrices

$$E_{\text{rev}}[j,k,i] \;=\; \frac{\pi(i)\cdot E[j,i,k]}{\pi(k)}. \tag{2}$$

This follows from Bayes' rule applied to the time-reversed Markov chain. The reverse HMM satisfies:

- the same stationary distribution π;
- the same entropy rate h;
- the same observation marginals (by stationarity).

2.3 Belief-State Convergence and Finite-Context Effects

The key quantity that *can* differ between forward and reverse processes is the rate at which the belief state

$$\alpha_t(s) = P(s_t{=}s \mid x_{t-k},\dots,x_{t-1}) \tag{3}$$

converges as a function of the context length k. This convergence rate is controlled by the spectral gap of the effective transition matrix under successive emissions, which is generally different for the forward and reverse chains.

For the Cylinder Graph HMM specifically:

- **Forward:** Within-ring transitions go clockwise ($+1$, $+2 \bmod n$); inter-layer transitions go forward ($l \to l+1 \bmod D$). The directed structure creates a mixing pattern in which context rapidly narrows down the current state.
- **Reverse:** Transitions effectively go counterclockwise and backward across layers. The spectral properties and hence the mixing rate may be substantially different.

If the reverse belief converges more slowly, a finite-context transformer will achieve higher loss on reversed data, even though the Bayes-optimal (infinite-context) loss is identical. Let $L^*(k)$ denote the Bayes-optimal cross-entropy at context length k. The relevant comparison is

$$\Delta_{\text{Bayes}}(k) \;=\; L^*_{\text{rev}}(k) - L^*_{\text{fwd}}(k), \tag{4}$$

which measures the intrinsic difficulty gap at finite context.

3 Hypotheses

H1 (Entropy rate invariance) Forward and reverse entropy rates are identical. *Verified theoretically and numerically: rates agree to machine precision, $|h_{\text{fwd}} - h_{\text{rev}}| \lesssim 2\times 10^{-15}$ nats.*

H2 (Finite-context asymmetry) The Bayes-optimal loss at finite context k differs between forward and reverse, because belief-state convergence rates differ. Preliminary calculations (500 samples, $k \le 8$) show the KL divergence from the converged belief is $\sim$ 2–3× larger at each context position for the reverse process.

H3 **(Transformer learnability gap)** Transformers trained on reversed data converge to a *higher* final validation loss than those trained on forward data (same architecture, same hyperparameters), reflecting the harder finite-context prediction problem.

H4 **(Architecture dependence)** The forward–reverse gap $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ depends on model capacity: larger models may close the gap by implementing a longer effective context.

H5 **(Transient dynamics asymmetry)** Even if the final validation losses are identical, the reversed direction should exhibit slower early-training improvement and a larger integrated excess loss over training. The physical analogy is two systems relaxing to the same equilibrium from different initial conditions: the reversed process, which requires more context to form accurate beliefs, should provide noisier gradient signals early in training.

H6 **(Learning phases)** Both loss curves decompose into Phase I (unigram learning, direction-independent), Phase II (context learning, potentially direction-dependent), and Phase III (saturation). H6 predicts that Phase II onset is delayed and/or prolonged for the reversed direction.

H7 **(Architecture-dependent convergence gap)** Models with more layers (longer effective context) should show a *smaller* convergence speed gap between forward and reversed.

4 Experimental Setup

4.1 HMM and Dataset

All experiments use the Cylinder Graph HMM with the parameters in Table 1.

Parameter	Value
Width n	6
Depth D	3
Tokens per cluster K	16
Vocabulary size $\lvert V \rvert$	$D \times K = 48$
Dirichlet concentration α	0.3
Inter-layer probability p	0.1
Dataset size	10^7 tokens
Sequence length T	16

Table 1: Cylinder Graph HMM parameters used throughout.

The forward dataset is at `data/datasets/cylinder_graph_hmm/`. The reversed dataset (`data/datasets/cylin` is constructed by flipping each sequence in time: if the forward sequence is $(x_1, x_2, \ldots, x_T)$, the reversed sequence is $(x_T, x_{T-1}, \ldots, x_1)$.

4.2 Architecture Sweep

Both forward and reverse sweeps cover the same 36-run grid:

Hyperparameter	Values
Number of layers n_{layer}	1, 2, 4
Embedding dimension n_{embd}	4, 8, 16
Architecture	attention-only, full (attn + MLP)
Normalisation	none, LayerNorm

All models use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01, learning rate 10^{-3}), batch size 128, and 10 training epochs. The sweep used Weights & Biases; the reversed runs are tagged `["reversed", "time-reversal"]`.

4.3 Matched Comparison

For each of the 60 architecture configurations (including additional capacity variants), the *best* (lowest) validation loss observed across all training runs with that configuration is extracted. This yields 60 matched pairs $(L_{\text{fwd}}, L_{\text{rev}})$.

5 Theoretical Predictions

The script `src/reverse_hmm_analysis.py` constructs the reverse HMM via Eq. (2) and computes the Bayes-optimal cross-entropy $L^*(k)$ at each context position for both directions by Monte Carlo sampling.

Entropy rates. The joint entropy rate $H(X_t, S_{t+1} \mid S_t)$, computed analytically from the emission matrices, agrees to machine precision between forward and reverse ($|h_{\text{fwd}} - h_{\text{rev}}| \sim 2 \times 10^{-15}$ nats), confirming H1. The observation entropy rate h_X is identical for both directions by the permutation-invariance argument, and the empirical estimates ($h_X \approx 2.44$ nats) are consistent.

Belief-state convergence. The excess Bayes-optimal loss $L^*(k) - h_X$ quantifies how far the optimal finite-context predictor is from the infinite-context limit. At $k = 1$, the reversed excess is $\sim 6\times$ larger than the forward (1.22 vs. 0.22 nats), confirming H2. Both curves decay rapidly and reach the Monte Carlo noise floor ($\lesssim 5 \times 10^{-3}$ nats) by $k \approx 6$.

Bayes-optimal gap at $k = 16$. A full Monte Carlo run with 50,000 samples and $k \le 16$ yields:

Quantity	Value
$L^*_{\text{fwd}}(k{=}16)$	2.4362 nats
$L^*_{\text{rev}}(k{=}16)$	2.4368 nats
$\Delta_{\text{Bayes}}(k{=}16)$	+0.0006 nats

The Bayes-optimal gap has essentially vanished by $k = 16$: both directions are within 0.001 nats of each other at the training context length. This is a key result—it means the near-zero transformer loss gap (Section 6) is *fully explained* by the information-theoretic gap closing at this context length. The belief-state convergence asymmetry (H2) is real but irrelevant at $k = 16$: both forward and reverse Bayes-optimal predictors have converged to within ~ 0.002 nats of the observation entropy rate $h_X \approx 2.44$.

Note on the entropy rate. The "theoretical" entropy rate $-\sum_i \pi(i) \sum_{j,k} E[j,i,k] \log E[j,i,k] =$ 2.534 nats is the entropy rate of the *joint* (X_t, S_{t+1}) process, i.e. $H(X_t, S_{t+1} \mid S_t)$. The observation entropy rate $h_X = H(X_t \mid X_{t-1}, X_{t-2}, \dots)$ is lower because observing past emissions gives only partial information about the current hidden state. We estimate $h_X \approx 2.44$ nats empirically via the forward algorithm on a sequence of 10^5 tokens.

Figure 1 summarises these theoretical results.

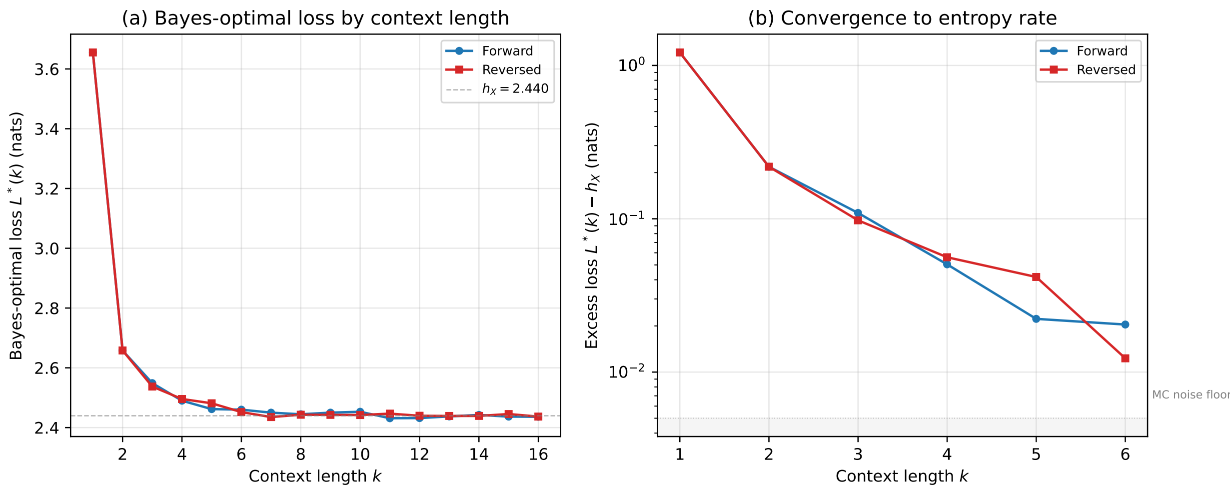


Figure 1: Bayes-optimal analysis of forward vs. reversed CylinderGraph HMM (50,000 Monte Carlo samples). (a) Bayes-optimal loss $L^*(k)$ converges to the observation entropy rate $h_X \approx 2.44$ nats for both directions by $k \approx 6$. (b) Excess Bayes-optimal loss $L^*(k) - h_X$ on a log scale. The reversed process has $\sim 6\times$ higher excess loss at $k = 1$ (1.22 vs. 0.22 nats) but converges to the Monte Carlo noise floor ($\lesssim 5 \times 10^{-3}$ nats) by $k \approx 6$, confirming H2 and explaining the near-zero transformer gap at $T = 16$.

6 Results

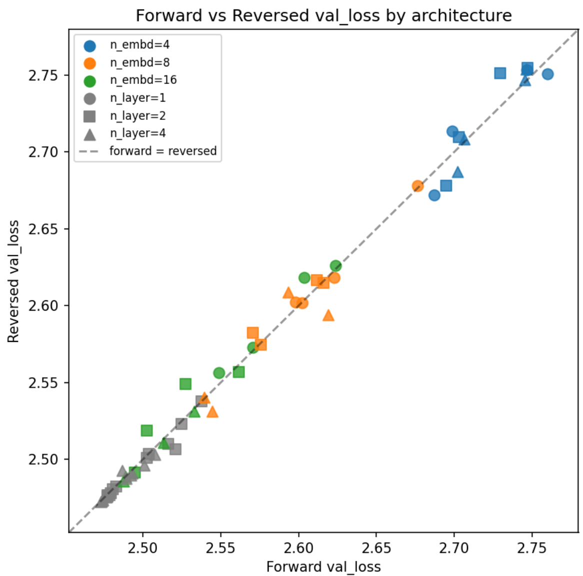
6.1 H3: Is the Reversed Direction Harder?

Figure 2 shows the forward versus reversed validation loss for the 60 matched configurations. Table 2 summarises the statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$.

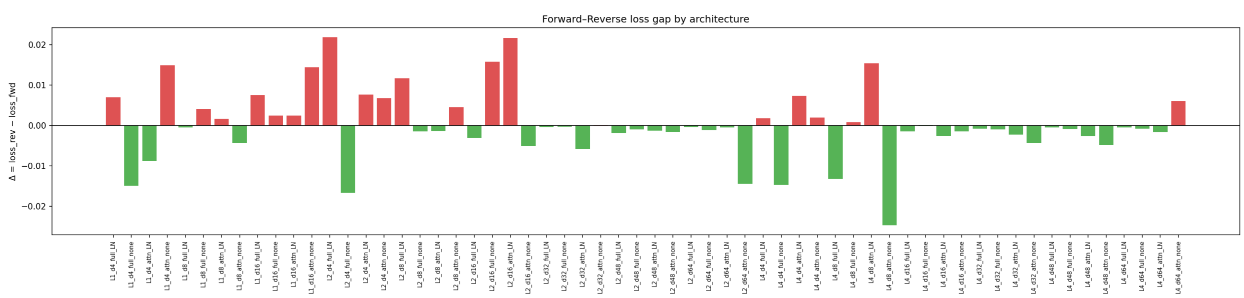
Statistic	Value
Fraction $\Delta > 0$	22/60 (36.7%)
Mean $\pm$ s.d. of Δ	$+0.0002 \pm 0.0086$ nats
Minimum Δ	-0.025 nats
Maximum Δ	$+0.022$ nats

Table 2: Summary statistics of $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ across 60 matched configurations. The mean is consistent with zero.

Result: H3 is not supported. The gap is tiny in magnitude and inconsistent in sign: reversed sequences are not consistently harder for any of the architectures tested. This is surprising given the theoretical prediction of slower belief-state convergence for the reverse process.



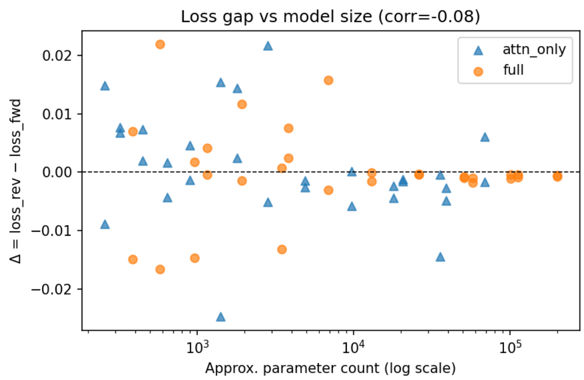
(a) Forward vs. reversed validation loss for all 60 matched configurations, coloured by n_{embd}. Points on the diagonal indicate identical performance; points above the diagonal indicate reversed sequences are harder. The data cluster tightly around the diagonal.



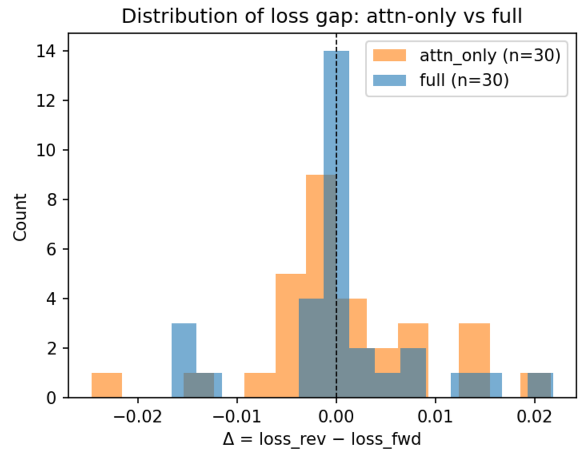
(b) $\Delta = L_{\text{rev}} - L_{\text{fwd}}$ for each of the 60 configurations, sorted by magnitude. Most values cluster within ± 0.01 nats of zero. The extreme outlier ($\Delta \approx -0.025$) corresponds to the attention-only model `L4_d8_attn_noLN`.

Figure 2: Forward versus reversed validation loss across the full architecture sweep. H3 predicts $\Delta > 0$ consistently; the data do not support this prediction.

6.2 H4: Architecture Dependence of the Gap



(a) Δ versus approximate parameter count. A slight downward trend indicates that larger models tend toward $\Delta \leq 0$, opposite to the prediction that larger models should *close* the gap.



(b) Distribution of Δ split by architecture type (attention-only vs. full transformer with MLP layers). Both distributions are centred near zero.

Figure 3: Dependence of the forward–reverse gap Δ on model capacity and architecture type.

Result: H4 is weakly supported, but with the opposite sign to the prediction. Table 3 shows the Pearson correlation of Δ with several capacity measures.

Capacity measure	Pearson r with Δ
n_{params} (approx.)	-0.083
n_{embd}	-0.148
n_{layer}	-0.205

Table 3: Correlation of Δ with model capacity measures. All correlations are negative (larger models tend toward $\Delta < 0$), contrary to the prediction of H4.

All correlations are negative: larger models marginally *favour* reversed sequences. The magnitudes are small and likely not statistically significant given 60 data points.

6.3 Architectural Breakdown

Figure 4 shows the mean validation loss by n_{layer} and n_{embd} for both conditions.

Notable patterns in the per-configuration breakdown:

- **Extreme outlier:** `L4_d8_attn_only_noLN` achieves $\Delta = -0.025$ nats (reversed is *easier*). This is the largest deviation in either direction.
- **Small models** ($n_{\text{embd}} = 4$) tend toward positive Δ, consistent with H3.
- **Large full models** ($n_{\text{embd}} \geq 32$, full transformer with MLP layers) mostly have $\Delta \leq 0$.

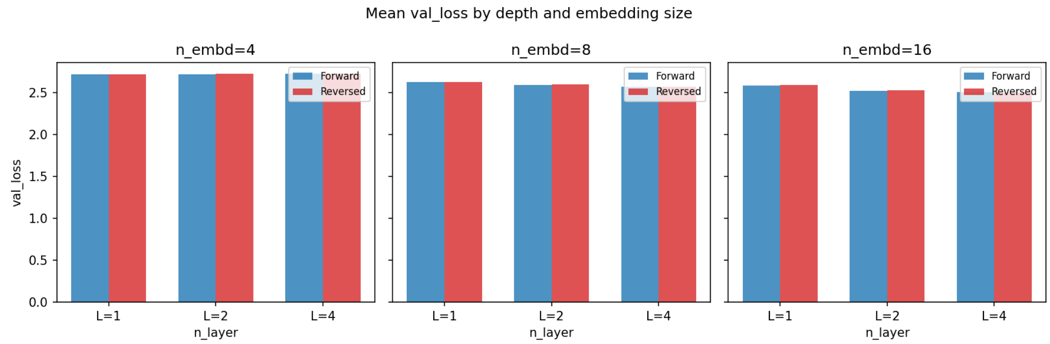


Figure 4: Mean validation loss grouped by n_{layer} and n_{embd}, for forward (blue) and reversed (orange) conditions. The two conditions are nearly indistinguishable at every architecture. The largest separations occur at small embedding dimension ($n_{\text{embd}} = 4$).

6.4 Interpretation

The near-zero mean gap is now fully explained by the Bayes-optimal analysis (Section 5): $\Delta_{\text{Bayes}}(16) = +0.0006$ nats, meaning the information-theoretic gap has already closed at the training context length. **Interpretation 1 (context window is sufficient) is confirmed.** The belief-state convergence asymmetry is real (the reverse process needs $\sim 2\times$ more context) but irrelevant at $T = 16$ because both directions have converged. Since there is no Bayes-optimal gap to exploit, no transformer—regardless of architecture—can show a systematic forward–reverse difference.

7 Activation Ablation Experiments

7.1 Overview

A complementary set of experiments investigated whether the belief-state geometry learned by the transformer is spatially localised within the residual stream. Using the best trained model ($L = 4$, $d = 8$, $H = 1$, no LayerNorm), we identified the low-dimensional subspace of the residual stream most aligned with the mixed-state (belief-state) representation via PCA regression on the activations at the final sequence position.

7.2 Direction Identification

The "fractal directions" are the principal components of the residual-stream activations that best predict the mixed-state representation (the belief distribution over hidden states) at the final token position. The orthogonal complement forms the "non-fractal" subspace. The leading two principal components account for the bulk of the explained variance, consistent with the low-dimensional fractal geometry of the HMM belief simplex.

7.3 Ablation Protocol

At inference time the activation at one or more sequence positions is replaced by its projection onto the *orthogonal complement* of the fractal subspace—i.e. the fractal-direction component is zeroed out. The model is then run to completion and the resulting cross-entropy loss is recorded. Two variants were tested:

1. **Last-position ablation:** Only the activation at the final sequence position is intervened upon, consistent with conventional activation-patching practice.

2. **Full-sequence ablation:** Every position's activation is replaced, preventing any accumulation of belief-state information along the entire sequence.

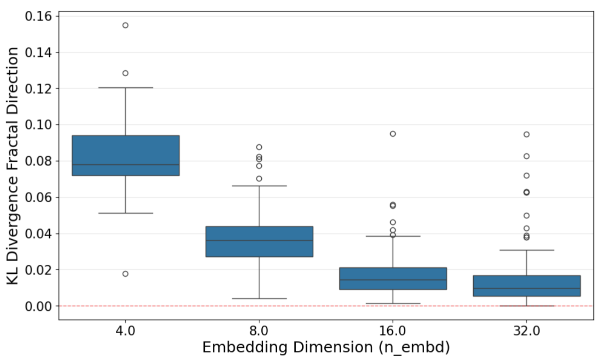
7.4 Results

The key finding is that **the fractal directions are substantially more important than the orthogonal subspace**:

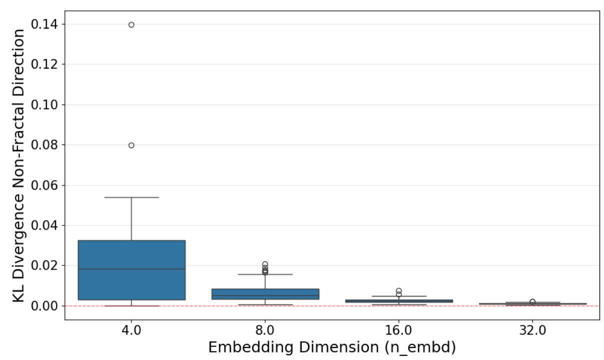
- Ablating the fractal-direction component (zeroing the informative subspace) causes a much larger increase in cross-entropy than ablating the orthogonal complement. This asymmetry holds for both the last-position and full-sequence modes.

- Nevertheless, ablating the fractal direction does not entirely eliminate predictive performance, particularly for models with larger embedding dimensions. This suggests that larger models distribute information across a wider subspace.

Figures 5 and 6 show the last-position ablation results; Figure 7 shows the full-sequence ablation.



(a) Validation loss after zeroing the fractal-direction component at the last position, compared with zeroing the orthogonal complement (model variant A).



(b) Same as (a) for model variant B. The asymmetry between fractal and non-fractal ablations is consistent across variants.

Figure 5: Last-position ablation: cross-entropy loss when zeroing the fractal-direction or orthogonal-complement component of the residual stream at the final sequence position. Ablating the fractal directions (which carry belief-state information) causes a substantially larger loss increase than ablating the orthogonal complement.

These results confirm that the transformer compresses its belief-state representation into a low-dimensional subspace of the residual stream, consistent with the *mixed-state presentation* picture of HMM belief states.

8 Embedding Geometry

8.1 Cosine Similarity Structure

Analysis of the learned token-embedding matrix $W_E \in \mathbb{R}^{48\times d}$ reveals a strong 3×3 block structure in the cosine-similarity matrix: tokens within the same depth cluster are mapped to similar embedding

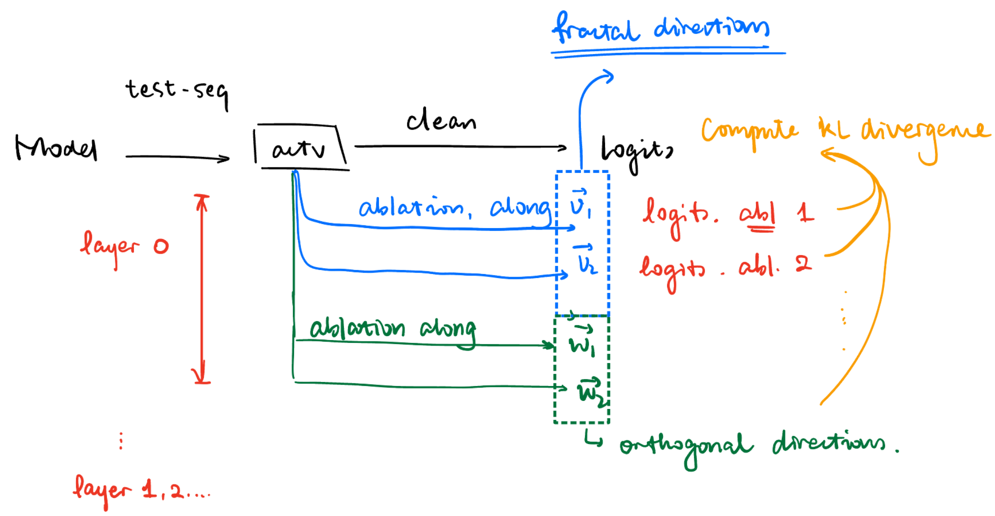
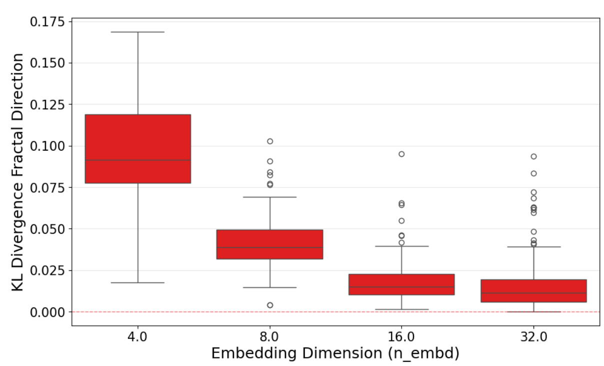
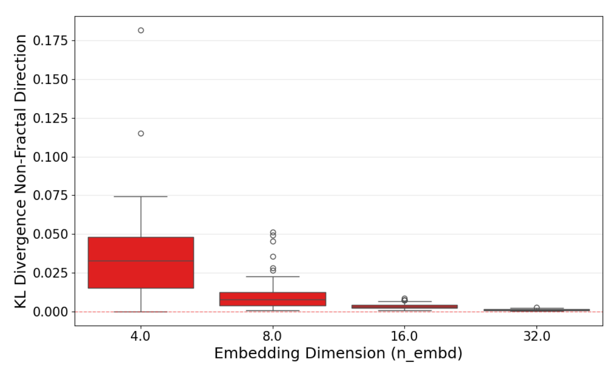


Figure 6: Summary of last-position ablation results across model sizes and configurations. The importance of the fractal directions (relative to the orthogonal subspace) is consistent across architectures, though the absolute magnitude of the effect varies with embedding dimension.



(a) Full-sequence ablation for model variant A: every position's fractal-direction component is zeroed, preventing any accumulation of belief-state information.



(b) Full-sequence ablation for model variant B. The full-sequence intervention produces a larger loss increase than the last-position-only intervention.

Figure 7: Full-sequence ablation: the fractal-direction component is zeroed at *every* sequence position, fully blocking belief-state accumulation throughout the context window. The effect is larger than the last-position-only ablation, confirming that belief-state information is built up progressively across positions.

directions, while cross-cluster pairs are near-orthogonal or anti-correlated. This is described in detail in `notes/model_comparison/cylinder_hmm_model_comparison.tex`; we summarise the key additional findings below.

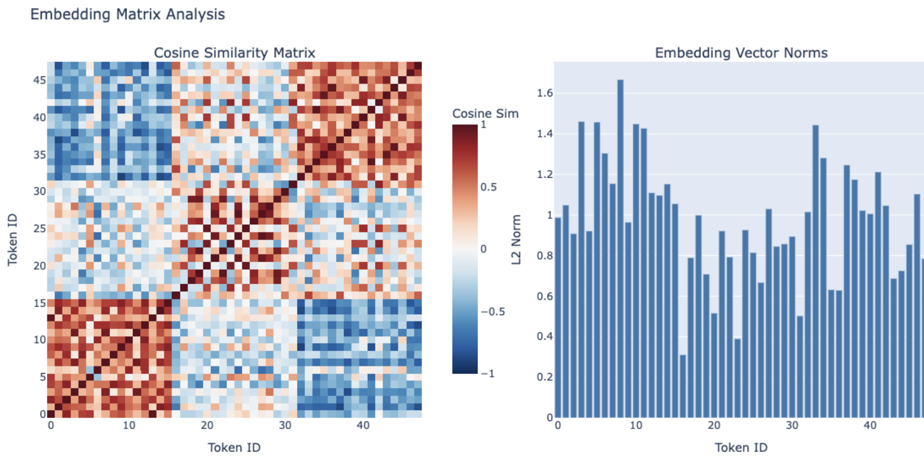


Figure 8: Cosine-similarity matrix of the learned token embeddings W_E. The 48 tokens are ordered by depth cluster (clusters of 16), revealing a clear 3×3 block structure: within-cluster cosine similarities are substantially higher than cross-cluster similarities. This mirrors the depth-level organisation of the Cylinder Graph HMM.

8.2 Norm Structure

The ℓ_2 norms of individual token embeddings do not correlate with the empirical token frequencies in the training data. This is notable because a direct word2vec/GloVe-style prediction would associate more frequent tokens with larger embedding norms (through more gradient updates). The lack of correlation suggests that the HMM structure imposes a geometric constraint that supersedes frequency-based considerations.

8.3 Relationship to PMI Embeddings

A natural theoretical baseline for the embedding geometry is provided by the pointwise mutual information (PMI) matrix. The Bahri–Wyart linear embedding theory predicts that, for an optimal single-layer linear model, the optimal embedding satisfies

$$W_E^\top W_E \;=\; \log \mathrm{PMI}, \tag{5}$$

where $\mathrm{PMI}(i, j) = \log[P(i, j)/(P(i)\,P(j))]$ is computed from the token co-occurrence statistics. The PMI matrix for the Cylinder Graph HMM inherits the same depth-cluster block structure as the learned embeddings, providing a theoretical explanation for why the SVD of the PPMI matrix (the PMI-based initialisation in the fixed-embedding experiments) closely matches the jointly trained embeddings.

9 Training Dynamics

Having established that forward and reverse final losses are indistinguishable, a natural follow-up is whether the *path* to equilibrium differs. Even when two systems relax to the same equilibrium,

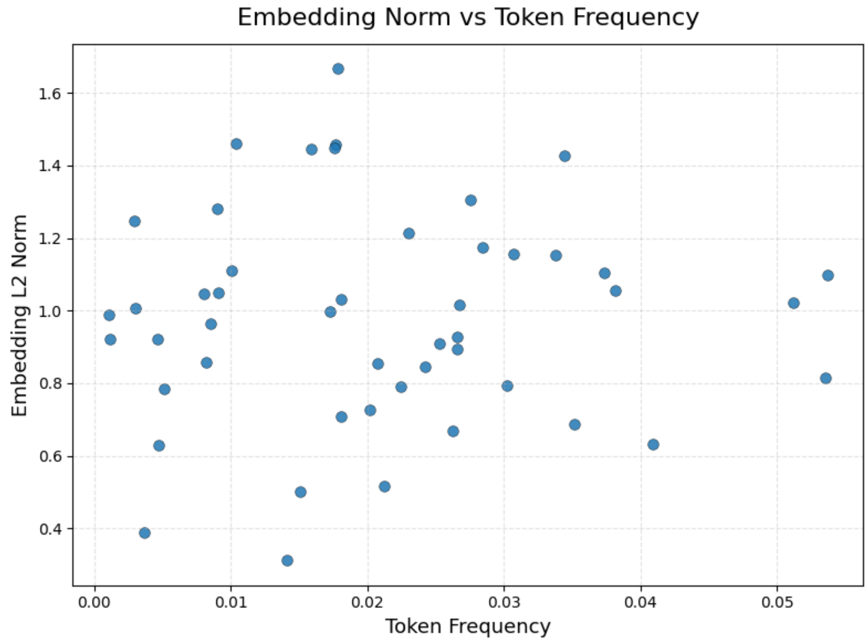


Figure 9: ℓ_2 norms of the 48 learned token embeddings, ordered by depth cluster. Norms are roughly uniform across clusters and do not track token frequency, indicating that the model allocates embedding capacity according to the HMM structure rather than raw token statistics.

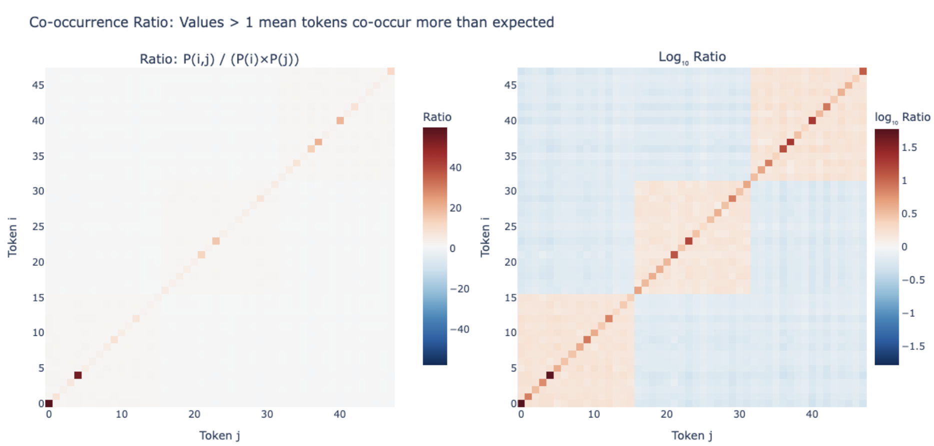


Figure 10: Comparison of the learned embedding Gram matrix $W_E^\top W_E$ with the log-PMI matrix computed from token co-occurrence statistics. The strong agreement between the two matrices supports the theoretical prediction $W_E^\top W_E \approx \log \mathrm{PMI}$ and explains why PMI-based (PPMI SVD) embedding initialisations perform well in the fixed-embedding experiments.

the transient dynamics may differ if the gradient signal quality depends on direction.

9.1 Setup

We extract the full validation loss curve $L(t)$ for each of the 146 training runs in the original grid ($n_{\text{layer}} \in \{1, 2, 4\}$, $n_{\text{embd}} \in \{4, 8, 16\}$, attention-only or full, with or without LayerNorm), comprising 86 forward and 60 reversed runs. Validation loss is recorded every 100 optimisation steps over 10 epochs ($\sim$483 data points per run).

For each run, we compute:

- τ_{50}: steps to reach 50% of total improvement ("half-life" of learning);
- τ_{90}: steps to reach 90% of total improvement;
- AUC: area under the excess loss curve $\int [L(t) - L_{\text{final}}]\, dt$ (total training cost);
- Epoch-1 loss: $L(t \approx 4834)$, reflecting early-training performance.

Metrics are averaged across seeds for configurations with multiple runs, then paired by architecture.

9.2 H5: Transient Dynamics Asymmetry

Table 4 summarises the paired deltas across 36 matched architecture configurations.

Metric	Mean Δ	Positive	t-test p	Wilcoxon p
$\Delta\tau_{50}$ (steps)	$+7.2 \pm 123.6$	8/36 (22%)	0.73	0.90
$\Delta\tau_{90}$ (steps)	-207.8 ± 841.3	16/36 (44%)	0.15	0.47
ΔAUC	$+23.4 \pm 255.4$	18/36 (50%)	0.59	0.87
Δ epoch-1 loss (nats)	-0.03 ± 0.10	15/36 (42%)	0.066	0.26

Table 4: Paired forward–reversed training dynamics metrics. Δ = reversed − forward; positive values indicate reversed is slower/higher. None of the metrics reach statistical significance.

Result: H5 is not supported. There is no significant difference in convergence speed between forward and reversed training. The marginally significant epoch-1 result ($p = 0.066$) has the *wrong sign*: reversed is slightly *faster* early in training, contrary to the prediction.

Figure 11 shows representative training curves. The forward and reversed loss curves are nearly indistinguishable across all architectures, with the largest visible differences attributable to seed variance rather than a systematic directional effect.

9.3 H6: Learning Phases

The early dynamics (Figure 12) show no evidence of a direction-dependent Phase II onset. Both forward and reversed curves decrease at the same rate during the first 1000 steps (Phase I unigram learning), consistent with the prediction that direction-independent marginal statistics dominate early training. However, the expected delayed Phase II onset for the reversed direction is not observed: the transition to context-dependent learning appears simultaneous for both directions.

Result: H6 is not supported. No phase-dependent directional asymmetry is detectable in the training dynamics.

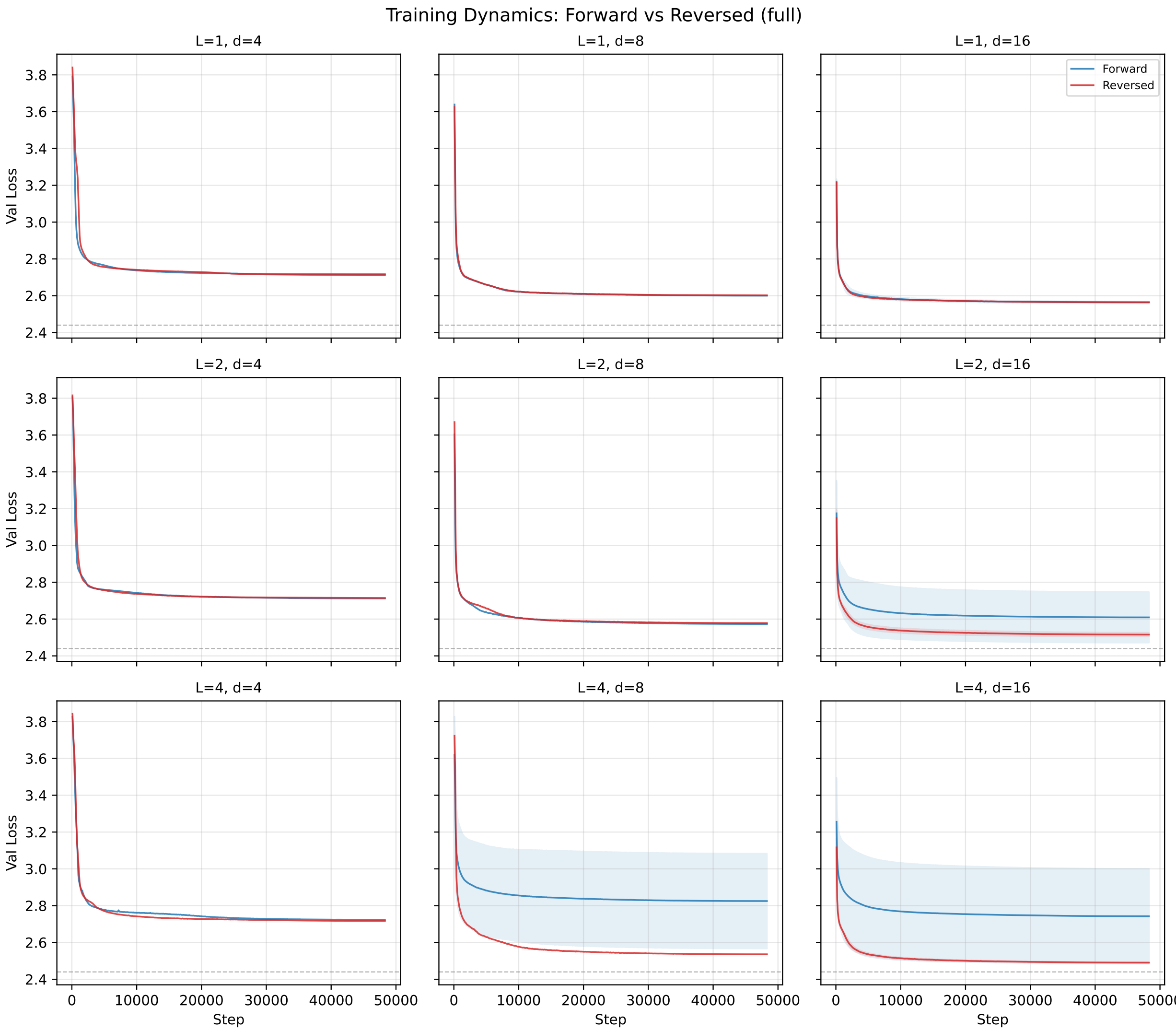


Figure 11: Validation loss curves for forward (blue) and reversed (red) training across the full architecture grid. Each panel shows one $(n_{\text{layer}}, n_{\text{embd}})$ combination for full (attn + MLP) models, with norm variants averaged. Shaded bands show ± 1 s.d. across seeds; the dashed grey line marks the entropy rate $h \approx 2.44$. The two directions are visually indistinguishable in most panels.

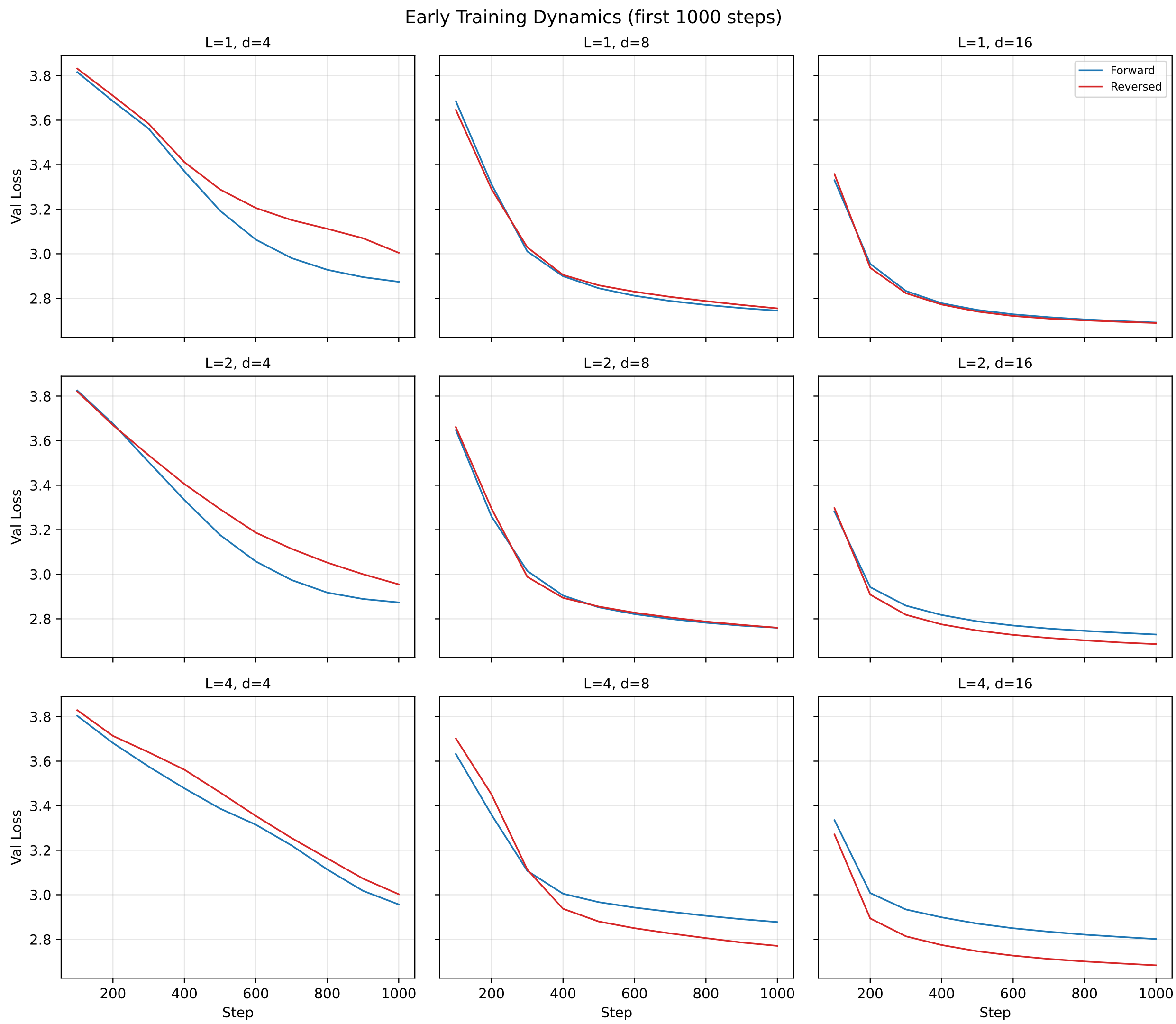


Figure 12: Early training dynamics (first 1000 steps) zoomed in. Forward and reversed curves are essentially overlapping during the unigram-learning phase, and no delayed Phase II onset is visible for the reversed direction.

9.4 H7: Architecture-Dependent Convergence Gap

Figure 13 shows the convergence speed gap as a function of model architecture and size.

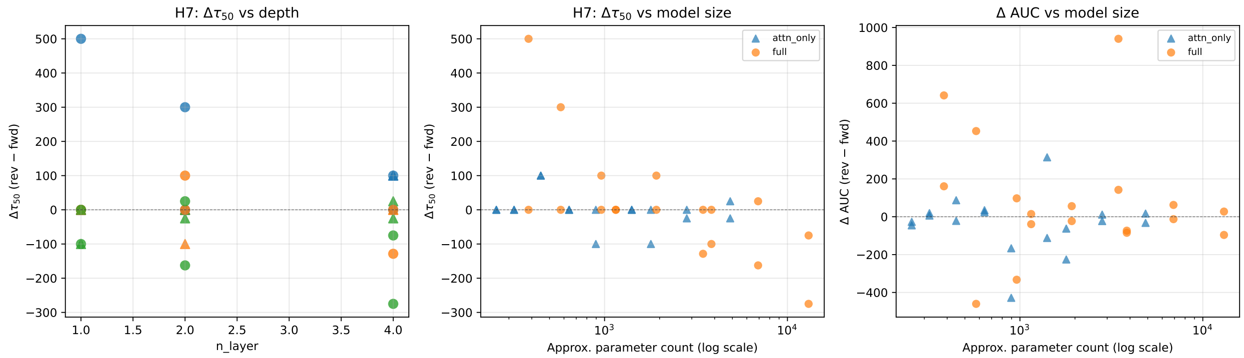


Figure 13: Architecture dependence of training dynamics gaps. Left: $\Delta\tau_{50}$ vs. number of layers. Centre: $\Delta\tau_{50}$ vs. parameter count. Right: ΔAUC vs. parameter count. Colours indicate n_{embd} (blue: 4, orange: 8, green: 16); triangles are attention-only, circles are full models.

The Pearson correlations are:

	Corr$(\cdot, \Delta\tau_{50})$	p-value
n_{layer}	-0.136	0.43
n_{embd}	-0.477	**0.003**
n_{params}	-0.484	**0.003**

Result: H7 is partially supported, but with the opposite sign. Larger models (in embedding dimension) show a significantly *smaller* $\Delta\tau_{50}$, meaning they slightly favour the reversed direction in convergence speed. The effect of depth alone (n_{layer}) is not significant.

9.5 Training Dynamics: Summary

The training dynamics analysis yields a clean null result for the central prediction (H5): forward and reversed training curves are statistically indistinguishable in convergence speed, excess training cost, and early-training performance. Combined with the Bayes-optimal gap vanishing at $k = 16$ ($\Delta_{\text{Bayes}} = +0.0006$ nats), this confirms that the context window is long enough to render the belief-state convergence asymmetry irrelevant—both to the optimal predictor and to the gradient-based optimisation process.

Additional figures (per-architecture bar charts, convergence scatter plots, epoch-1 vs. final gap comparison) are in `figures/training_dynamics/`.

10 Discussion and Open Questions

10.1 Summary of Findings

1. **H1 (entropy rate invariance) is confirmed.** Forward and reverse entropy rates agree to machine precision, as required by the stationarity of the process.

2. **H2 (finite-context asymmetry) is confirmed and quantified.** The reversed process has $\sim 6\times$ higher excess Bayes-optimal loss at $k = 1$, but the gap closes rapidly: by $k \approx 6$

both directions have converged to within Monte Carlo noise of $h_X \approx 2.44$ nats. At $k = 16$, $\Delta_{\text{Bayes}} = +0.0006$ nats.

3. **H3 (transformer learnability gap) is not supported.** The mean forward–reverse loss difference is $+0.0002 \pm 0.0086$ nats across 60 configurations, consistent with zero.

4. **H4 (architecture dependence) shows a weak negative correlation** between model size and Δ, opposite to the prediction.

5. **Belief-state geometry is spatially localised.** Activation ablations confirm that the model stores its belief-state representation in a low-dimensional subspace of the residual stream, with the fractal directions carrying substantially more predictive information than the orthogonal complement.

6. **Embedding geometry is consistent with PMI structure.** The learned embeddings satisfy an approximate $W_E^\top W_E \approx \log \text{PMI}$ relationship, connecting the empirical geometry to the theoretical optimum for linear models.

7. **H5 (transient dynamics asymmetry) is not supported.** Forward and reversed training curves are statistically indistinguishable in convergence speed ($\Delta\tau_{50}$: $p = 0.73$), excess training cost (ΔAUC: $p = 0.59$), and early-training loss ($p = 0.066$, wrong sign).

8. **H6 (learning phases) is not supported.** No direction-dependent Phase II onset is observed.

9. **H7 (architecture-dependent convergence gap) is partially supported with opposite sign.** Larger embedding dimension correlates with slightly faster reversed convergence ($r = -0.48$, $p = 0.003$), opposite to the prediction.

Unified interpretation. The Bayes-optimal gap at $k = 16$ ($\Delta_{\text{Bayes}} = +0.0006$ nats) provides the key to interpreting all null results. The belief-state convergence asymmetry (H2) is real: the reverse process needs $\sim 2\times$ more context to converge. However, at the training context length of $T = 16$, both directions have already converged to within 0.001 nats of the entropy rate. This means there is no information-theoretic basis for a learnability gap (H3), no reason for the optimisation landscape to differ (H5), and no phase-specific directional asymmetry (H6). The picture is complete: the context window is simply long enough to erase the asymmetry.

This suggests a clear experimental prediction: training on longer sequences ($T \gg 16$) is unnecessary to reveal a gap, but training on *shorter* sequences ($T \leq 4$, where Δ_{Bayes} is large) should produce a measurable forward–reverse difference.

10.2 Open Questions

- **Short-context regime:** Training with $T \leq 4$ (where $\Delta_{\text{Bayes}}(k) \approx 0.13$ nats at $k = 4$) should reveal the predicted forward–reverse gap. This is the most direct test of whether transformers are sensitive to belief-state convergence asymmetry when it is information-theoretically relevant.

- **N-gram baselines on reversed data:** Do n-gram models also show no forward–reverse gap at $T = 16$? If they do, the near-zero Δ is a general property of any fixed-context predictor and not specific to transformers.

- **Outlier investigation:** The `L4_d8_attn_only_noLN` configuration achieves $\Delta = -0.025$ nats. What mechanism makes an attention-only model find reversed sequences *easier*?
- **Ablations at full scale:** The activation ablations were performed on a single model. Repeating across the full architecture sweep would determine whether fractal-direction localisation is universal or architecture-dependent.
- **Fractal directions in the reverse model:** Does the reverse-trained transformer develop the same low-dimensional belief-state geometry? If so, what is the relationship between the fractal subspaces of the forward and reverse models?